

Tirumala Venkata Durga Prasad Pitani

AI Engineer — Agentic AI — Generative AI — Enterprise Automation

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Professional Summary

I am an AI Engineer focused on turning foundation models into reliable enterprise automations. My work centers on designing agentic planning architectures, advanced RAG pipelines, and multi-agent workflows that solve complex business problems. I enjoy exploring fine-tuning strategies to adapt models to specific domains, and I build robust, scalable solutions across major cloud environments—primarily Azure, with complementary experience in GCP, AWS, and Snowflake. I care deeply about bridging the gap between experimental AI and production reality through explicit evaluation, human-in-the-loop design, and maintainable software engineering.

Technical Skills

- **Agentic AI & LLMs:** Multi-Agent Orchestration, ReAct & Tool-Use Agents, LangChain, LangGraph, LlamaIndex, Prompt Engineering, Fine-Tuning Exploration, Structured Outputs
- **RAG & Knowledge Retrieval:** Advanced Vector Search (FAISS, ChromaDB, Pinecone), Semantic Chunking, Hybrid Retrieval, Context Grounding, NL-to-SQL
- **Cloud & Data Platforms:** Microsoft Azure (Primary), GCP, AWS, Snowflake, Azure Fabric, Databricks
- **Software Engineering:** Python (Async), SQL, FastAPI, Django REST, Docker, CI/CD, Automated Testing
- **MLOps & Applied ML:** MLflow, Scikit-Learn, XGBoost, Model Evaluation, Deployment Pipelines

Professional Experience

Data Engineer / AI Engineer

NTT DATA Business Solutions

October 2024 – Present

Bengaluru, India

Primary Tools: Azure OpenAI, LangChain, FAISS, Azure Fabric, PySpark, Azure Data Factory

- Designed and deployed an LLM-powered enterprise analytics copilot, enabling secure natural-language queries against governed lakehouse data to streamline business intelligence workflows.
- Developed reusable AI components for prompt construction, retrieval, SQL generation, and validation, accelerating the adoption of maintainable multi-agent workflows across the enterprise.
- Built AI-ready data infrastructure on Azure Fabric, processing daily supply-chain data through Medallion Architecture and optimizing Delta Lake workloads for downstream RAG consumption.
- Engineered automated ETL pipelines in Azure Data Factory with dynamic parameterization and data-quality validation to ensure reliable reporting and grounding data.

Data Science Intern

Viai Solutions Pvt Ltd

July 2024 – August 2024

Hyderabad, India

Primary Tools: Python, PySpark, Selenium, BeautifulSoup, Pandas

- Developed automated Python data-ingestion workflows to extract and transform unstructured web data into Parquet datasets, significantly reducing manual collection effort.
- Processed and validated over 100K student records using PySpark, establishing rigorous data quality rules that dramatically improved dataset reliability.

Python Developer

Dhanush Infotech Pvt Ltd

January 2024 – June 2024

Hyderabad, India

Primary Tools: Python, Django REST Framework, SQL, JWT

- Developed scalable microservices with JWT authentication and role-based access control, supporting thousands of daily API requests with high availability.
- Optimized SQL schemas and query execution plans, improving API response times and reducing database CPU utilization.

Key Projects

Contract Risk Analyzer

github.com/Tirumala2824/contract-risk-analyzer

Primary Tools: FastAPI, Multi-Agent Orchestration, RAG, MongoDB, Python

- Developed an explainable contract-review automation that replaces black-box classification with an agentic planning architecture. The system orchestrates category-specific agents (Legal, Financial, Security) to analyze documents against enterprise policies.
- Engineered a multi-judge scoring engine combining rule-based, template, Bayesian, and LLM judges to calculate confidence-aware risk scores, demonstrating practical human-in-the-loop AI product design.

Cross-Sell Prediction System

github.com/Tirumala2824/Cross-sell-prediction

Primary Tools: Python, Scikit-Learn, Pandas, MLflow

- Engineered an applied machine learning pipeline to predict customer cross-sell opportunities, evaluating Logistic Regression against Support Vector Machines using ROC-AUC metrics.
- Focused on reusable data preprocessing and reproducible model training, bridging the gap between raw customer behavior data and actionable marketing targets.

Achievements & Certifications

- **1st Place** – Business AI Challenge 2026 (Cohort 1); **3rd Position** – JouleXet 2025 AI Hackathon
- **Certifications:** Microsoft Certified: Fabric Data Engineer Associate (DP-700), Azure AI Fundamentals (AI-900)
- **Publication:** "CNN-based Industrial Defect Detection System" (94% detection accuracy)

Education

B.Tech (Computer Science and Engineering)

IIIT Manipur

2020–2024